

05 — Tech, Libraries & Dependencies

What's installed, what each thing is doing, and why it was picked.

Runtime requirements

Component	Version	Why
Python	3.11	Match Dockerfile; needed for <code>from __future__ import annotations</code> style + newer typing syntax used in <code>store.py</code> .
Node.js	22 (LTS)	Vite 8 requires Node 20.19+ or 22.12+. Edge box uses Node 22 in the frontend container.
SQLite	bundled with Python	No external server. WAL mode for concurrent reads.
Disk	~2 GB	sentence-transformers wheels + Chroma index + uploaded PDFs grow over time.
RAM	~2 GB	SentenceTransformer resident set, embeddings during ingestion.

Python dependencies (requirements.txt)

```
docling
langchain
langchain-text-splitters
langchain-openai
langchain-community
qdrant-client
fastapi>=0.111.0
uvicorn[standard]>=0.30.0
streamlit
python-dotenv>=1.0.0
chromadb>=0.5.0
sentence-transformers>=3.0.0
groq>=0.9.0
resend>=2.0.0
pypdf>=4.0.0
pydantic[email]>=2.0.0
python-multipart>=0.0.9
email-validator>=2.0.0
```

Used at runtime

Package	Where	Why
<code>fastapi</code>	<code>src/api.py</code>	HTTP framework. Pydantic models for request/response shapes.
<code>uvicorn[standard]</code>	entrypoint	ASGI server. <code>[standard]</code> pulls <code>httptools</code> + <code>uvloop</code> for perf.
<code>python-dotenv</code>	<code>src/api.py</code> (<code>load_dotenv()</code>)	Reads <code>.env</code> at boot — <code>GROQ_API_KEY</code> , <code>RESEND_API_KEY</code> , <code>ADMIN_EMAILS</code> , <code>APP_BASE_URL</code> , <code>ALERT_THRESHOLD</code> , <code>ALERT_DEDUP_SECONDS</code> .
<code>chromadb</code>	<code>src/db.py</code> , <code>src/api.py</code>	Local vector index with cosine similarity. Persistent client at <code>./chroma_db</code> .
<code>sentence-transformers</code>	<code>src/api.py</code> lifespan + retriever	Loads <code>all-MiniLM-L6-v2</code> (384-dim embeddings) in-process. ~90 MB model.
<code>groq</code>	<code>src/llm_client.py</code>	Hosted Llama-3.1-70B for the completion step. Lazy singleton client.
<code>resend</code>	<code>src/mailer.py</code>	Transactional email — magic-link + alert emails.
<code>docling</code>	<code>src/ingestion/parser_chunker.py</code>	PDF → structured chunks. Replaces a raw <code>pypdf</code> approach.
<code>pypdf</code>	indirect, via docling	PDF text extraction fallback.
<code>pydantic[email]</code>	request models	Validation. <code>[email]</code> for <code>EmailStr</code> .
<code>python-multipart</code>	<code>POST /admin/machines</code>	Needed by FastAPI to parse multipart form uploads.
<code>email-validator</code>	indirect	Backs <code>EmailStr</code> .
<code>langchain-text-splitters</code>	ingestion	Recursive character splitter for fallback chunking.

Currently present but underused

Package	Status
langchain , langchain-openai , langchain-community	Imported in places but not load-bearing. Could be trimmed once ingestion is rewritten or wrapped in optional extras.
qdrant-client	Not used. Earlier exploration of Qdrant before settling on Chroma. Safe to remove.
streamlit	Not used in production. Was an early prototype UI. Safe to remove.

A cleanup PR could drop qdrant-client + streamlit + the three langchain-* packages if no one's relying on them — saves ~150 MB on the docker image.

Frontend dependencies (frontend/package.json)

```
"dependencies": {
  "@tailwindcss/typography": "^0.5.19",
  "autoprefixer": "^10.5.0",
  "framer-motion": "^12.38.0",
  "lucide-react": "^1.8.0",
  "postcss": "^8.5.10",
  "react": "^19.2.5",
  "react-dom": "^19.2.5",
  "react-markdown": "^10.1.0",
  "react-router-dom": "^7.14.1",
  "remark-gfm": "^4.0.1",
  "tailwindcss": "^3.4.19"
}
```

Package	Where	Why
react + react-dom	everywhere	Core framework. v19.
react-router-dom	App.jsx	Client-side routing. v7.
framer-motion	Navbar , MachineCard , LandingPage , modals	All animations + scroll-hijack carousel + layout transitions.
tailwindcss + postcss + autoprefixer	tailwind.config.js	Utility-first CSS. Most styling is inline className .
@tailwindcss/typography	for the prose class	Used on marketing pages + chat answer rendering.
lucide-react	every component	Icons. Tree-shakable; only the imported icons bundle.
react-markdown + remark-gfm	MessageContent	Renders LLM answers with GFM (tables, strikethrough, task lists).

Dev dependencies

Package	Purpose
<code>vite</code>	Dev server + build. <code>vite build</code> outputs to <code>frontend/dist/</code> .
<code>@vitejs/plugin-react</code>	React + JSX support.
<code>eslint</code> + <code>eslint-plugin-react-hooks</code> + <code>eslint-plugin-react-refresh</code>	Linting. <code>npm run lint</code> .
<code>@types/react</code> + <code>@types/react-dom</code>	Type hints for IDE / TS-aware linters. (Project is JSX, not TS.)
<code>globals</code>	ESLint global declarations.

External services

Service	Used for	Failure mode
Groq	LLM completion in <code>/query</code>	Returns <code>status: "error"</code> to the chat; doesn't crash. Workers see an error message.
Resend	Magic-link sign-in + alert emails	Fire-and-forget — alert paths log + persist record with <code>email_notified: false</code> . Magic-link uses an Resend API call directly — failure returns 502 to the admin.

Both are cloud services. **The edge box needs outbound HTTPS** for normal operation. Air-gap viable for: machine list, shift logging, parameter editing, analytics — anything that doesn't hit `/query`.

Required `.env` keys

```
GROQ_API_KEY=...
RESEND_API_KEY=re_...
ADMIN_EMAILS=alice@tecdia.com,bob@tecdia.com # comma-separated allowlist
APP_BASE_URL=http://localhost:5173 # what gets baked into magic-link URLs
# Optional
MAIL_FROM=Tecdia SmartFix <admin@your-verified-domain.com>
ALERT_THRESHOLD=12
ALERT_DEDUP_SECONDS=300
STORE_PATH=./smartfix.db # SQLite location; auto-set in compose
```

See `.env.example` for the full annotated template.

Build + deploy tooling

File	Role
Dockerfile	Backend image (Python 3.11 slim + uvicorn).
frontend/Dockerfile	Two-stage frontend image (Node 22 build → nginx alpine runtime).
frontend/nginx.conf	Mirrors Vite dev proxy so prod and dev behave the same.
docker-compose.yml	Wires both services + three named volumes (chroma_db , data , sqlite_store).
scripts/build_index.py	Offline ingestion (build Chroma from local JSONL chunks).
scripts/backup_sqlite.sh	WAL-safe nightly SQLite backup with 14-day rotation.
scripts/demo.py	Self-contained demo without real PDFs.
postman/SmartFix.postman_collection.json	API testing collection (current as of the v2 contract; may need refresh for the new admin routes).

Versions snapshot

These are the versions that were running as of the last handover. If you need to upgrade, do it one at a time and re-run `npm run build + pytest` (when tests exist).

Layer	Version
Python	3.11.x
FastAPI	≥ 0.111
ChromaDB	≥ 0.5
sentence-transformers	≥ 3.0
Groq SDK	≥ 0.9
Resend SDK	≥ 2.0
Node	22
Vite	^8.0.9
React	^19.2.5
framer-motion	^12.38.0
tailwindcss	^3.4.19